

Your grade: 100%

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Next item →

1. Which of the following are true about the discriminator in a GAN model?

1 / 1 point

- ☐ It generates new data samples from random noise
- ☒ It is updated based on the error it makes in classifying samples

 **Correct**
Correct! The discriminator is updated based on the error in its classification of real vs. generated samples.

- ☐ It combines multiple data sources into one
- ☒ It distinguishes between real and generated data samples

 **Correct**
Correct! The discriminator's role is to distinguish between real and generated data samples.

- ☒ It is trained to minimize a loss function

 **Correct**
Good job! The discriminator is trained to minimize a loss function that measures its ability to distinguish between real and fake samples.

2. What is the primary function of the generator in a GAN model?

1 / 1 point

- ☐ To reduce the dimensionality of data samples
- ☐ To combine multiple data samples into one
- ☒ To generate realistic data samples from random noise
- ☐ To classify data samples into different categories

 **Correct**
Correct! The generator's main goal is to create realistic data samples from random noise input.

3. What is the primary purpose of compiling the combined GAN model?

1 / 1 point

- ☐ To initialize the weights of both generator and discriminator.
- ☒ To train the generator to produce realistic data.
- ☐ To optimize the discriminator's ability to detect fake data.
- ☐ To balance the training speeds of the generator and the discriminator.

 **Correct**
Correct! The primary goal is to train the generator so that it can produce realistic data that can fool the discriminator.

4. Which method is used to train a model with a batch of images or data in GANs?

1 / 1 point

- ☒ train_on_batch
- ☐ compile_batch
- ☐ fit_on_batch
- ☐ evaluate_batch

 **Correct**
Correct! The train_on_batch method is used to train a model with a batch of data.

5. What are some key activities involved in training the discriminator in a GAN?

1 / 1 point

- ☒ Providing the discriminator with fake images generated by the generator

 **Correct**
Correct! This is a fundamental part of training the discriminator.

- ☒ Feeding the discriminator real images with corresponding labels

 **Correct**
Correct! Real images and their labels are used to train the discriminator.

- ☐ Updating the generator's weights directly

- ☒ Calculating the loss based on the discriminator's predictions

 **Correct**
Correct! The loss calculation is crucial for updating the discriminator's weights.

- ☐ Using a validation dataset to tune hyperparameters